

Project Title

Anomaly Detection in Industrial Images Using Diverse Deep Learning Backbones: A Study on MVTec-AD

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Abstract

Anomaly detection is essential in industrial inspection tasks to maintain product quality and minimize downtime. In this study, we build upon the PaDiM framework proposed by Defard et al. (2021), which models patch-level features using multivariate Gaussians and scores anomalies via Mahalanobis distance. Our work extends this approach by comparing the performance of four different deep learning backbones (ResNet18, ResNet50, DenseNet121, and EfficientNet-B5) on the MVTec-AD dataset. We incorporate several enhancements, including random channel selection, category-specific PCA threshold tuning, and feature extraction layer selection, to evaluate their impact on detection accuracy. Performance is measured using AUROC scores and confusion matrices. Our results show that EfficientNet-B5 achieves superior performance in most categories, particularly when synthetic anomalies are used to mitigate class imbalance. This study also highlights the importance of architecture depth, fine-tuned preprocessing, and dimensionality control in building robust unsupervised anomaly detection systems for industrial applications.

1. Introduction

Anomaly detection is the process of identifying patterns or observations that deviate significantly from what is considered normal. In visual inspection tasks, these anomalies typically represent manufacturing defects, damage, or irregularities that may compromise product quality or operational safety. The ability to automatically detect such anomalies has become increasingly important across various industries, including manufacturing, oil and gas, healthcare, and infrastructure.

In industrial environments, particularly those transitioning toward Industry 4.0 standards, the demand for accurate and scalable inspection systems has grown rapidly. Traditional manual inspection techniques often fail to meet the required levels of precision and speed, while rule-based systems lack the adaptability to handle real-world complexity and unforeseen defect types.

Deep learning offers a robust alternative by enabling automated feature learning directly from image data. In unsupervised anomaly detection, where only non-defective samples are available during training, convolutional neural networks (CNNs) have demonstrated strong capabilities in identifying unexpected patterns without explicit defect labels.

This project builds upon the PaDiM framework (Defard et al., 2021), which models patch-level feature distributions using multivariate Gaussians and scores anomalies via Mahalanobis distance. Our focus is on evaluating how different deep learning backbone architectures specifically ResNet18, ResNet50, DenseNet121, and EfficientNet-B5 affect anomaly detection performance on the MVTec-AD dataset. We also explore the effects of channel selection strategies, PCA dimensionality tuning, and synthetic data generation for underrepresented categories.

Through this comparative study, we aim to provide insight into how architectural design and preprocessing decisions influence the detection accuracy, generalizability, and robustness of unsupervised visual anomaly detection systems.

Overall Goal:

To evaluate the effectiveness of different CNN backbones (ResNet18, ResNet50, DenseNet121, EfficientNet-B5) in unsupervised image anomaly detection using the MVTec-AD dataset, with focus on real-world industrial applications.

Objectives

- Use and fine-tune popular CNN backbones (i.e, ResNet18, ResNet50, DenseNet121, EfficientNet-B5) as feature extractors for anomaly detection.
- Build a modular detection pipeline that handles patch-level feature extraction, random channel selection, optional PCA, and Gaussian modeling.
- Compare models using metrics like AUROC, precision, recall, F1-score, and confusion matrix at both image and pixel level.
- Try different layer combinations and depths for each category (e.g., screw, wood, transistor) and find what works best.

- Run tests with and without PCA, try different channel settings and layers to see how each part affects accuracy and speed.
- Check model size, processing speed, and GPU needs to see which ones are practical for use in real-time or low-resource setups.

Dataset Description

Dataset: MVTec Anomaly Detection (MVTec-AD) dataset is a comprehensive, publicly available benchmark designed specifically for research in unsupervised anomaly detection and localization in visual inspection tasks. Published by Bergmann et al. (2019), it replicates real industrial scenarios, offering high-res images of manufactured objects and textures each with authentic defect variations and precise ground truth annotations. Covers 15 different categories (10 objects, 5 textures), representing a range of products found in advanced manufacturing, packaging, and quality control.

Total Categories: 15 (e.g., bottle, hazelnut, transistor, toothbrush, screw, pill, tile, metal nut, leather, hazelnut, grid, capsule, cable, carpet, wood and zipper.)

Image Types: High-resolution RGB

Classes:

- good: normal images
- defect: various defect types (cracks, scratches, missing parts)

Preprocessing:

- All images resized to a standard size
- Converted to tensors and normalized using ImageNet statistics

2. Literature Review

2.1 Anomaly Detection

Anomaly detection has gained wide attention in computer vision, particularly in industrial quality control where detecting rare or subtle defects is critical. Traditional methods such as k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), Gaussian Mixture Models (GMM), and Local Binary Patterns (LBP) relied on handcrafted features like Gabor filters or texture statistics (Chandola et al., 2009; Liu et al., 2020). However, these methods often struggle to generalize to high-variability or unseen conditions, especially in real-world settings.

Recent advances have shifted towards deep learning, which enables automatic feature extraction from raw image data. Among these, unsupervised methods have become popular in industrial applications where defective samples are rare or unlabeled. These include autoencoders (Bergmann et al., 2019), generative models like GANomaly (Akçay et al., 2018), and distributional methods like PaDiM (Defard et al., 2021), which model only the normal class and detect deviations during inference.

2.2 Patch-Based Statistical Modeling Approaches

Among unsupervised approaches, **patch-based methods** like PaDiM (Defard et al., 2021) have become state-of-the-art. PaDiM leverages features from pre-trained CNN backbones and models the distribution of patch-level embeddings using multivariate Gaussians. At test time, it uses **Mahalanobis distance** to compare each test patch to the learned distribution of normal patches. This method achieves both high accuracy and strong localization without requiring defect labels during training.

Other good methods includes:

- **SPADE** (Cohen & Hoshen, 2020): Uses nearest-neighbor distances in feature space.
- **PatchCore** (Roth et al., 2022): Combines feature sampling and coreset selection to model normal patch distributions efficiently.
- **DRAEM** (Zavrtanik et al., 2021): Uses image reconstruction with pixel-wise training on synthetic anomalies.
- **GLASS** (Yang et al., 2023): Combines global semantic and local structural features to detect anomalies with semantic consistency, improving performance on complex and small defect cases

While these methods vary based on implementation, they often depend on the quality of the backbone features used in the pipeline making backbone selection a key focus of this project.

2.3 Deep Backbone Architectures for Feature Extraction

Pre-trained CNNs are commonly used as feature extractors in anomaly detection pipelines. The choice of backbone significantly influences the expressiveness and discriminative power of the extracted features. The most widely used architectures include:

- **ResNet** (He et al., 2016): Introduces residual connections to enable training of very deep networks. ResNet18 and ResNet50 are commonly used due to their simplicity and speed.
- **DenseNet** (Huang et al., 2017): Connects each layer to every other layer in a feed-forward manner, encouraging feature reuse and improving gradient flow. DenseNet121 is particularly effective for fine-grained anomaly localization.
- **EfficientNet** (Tan & Le, 2019): Uses compound scaling to balance depth, width, and resolution, achieving state-of-the-art performance with fewer parameters. EfficientNet-B5 offers a good trade-off between model size and accuracy.

Despite their success, most existing literature uses only one backbone per method. This project aims to systematically compare multiple backbones within the same anomaly detection framework to understand their individual trade-offs.

2.4 Dimensionality Reduction and Statistical Scoring

To reduce the high dimensionality of feature embeddings, techniques like Principal Component Analysis (PCA) are commonly used (Jolliffe, 2002). PCA helps eliminate redundant information, reduces overfitting, and improves numerical stability for covariance matrix estimation. In PaDiM-based approaches (Defard et al., 2021), patch features are projected into a lower-dimensional space, and a Gaussian distribution is fitted per spatial location. The Mahalanobis distance is then used to compute anomaly scores for each patch (De Maesschalck et al., 2000). This statistical scoring method is robust to scale and correlation differences in the data.

2.5 Benchmark Dataset: MVTec-AD

The MVTec Anomaly Detection (MVTec-AD) dataset (Bergmann et al., 2019) is a widely used benchmark for visual anomaly detection. It contains 15 industrial categories (e.g., capsule, transistor, carpet, wood), each with high-resolution RGB images of good and defective samples. Each defective image includes a pixel-level ground truth mask for localization. Recent releases such as MVTec AD 2 (Bergmann et al., 2023) introduce greater challenges including small defects, reflective surfaces, and complex lighting — encouraging more robust and adaptive detection methods.

2.6 Gaps and Motivation for This Work

While PaDiM and similar methods have shown strong performance, most implementations use only a single backbone model and limited pipeline tuning. There is limited literature on how backbone architecture, layer selection, channel sampling, and PCA tuning affect performance across different categories. Moreover, statistical significance testing is rarely applied when comparing model performance.

This project addresses these gaps by:

- Benchmarking multiple backbone networks in a controlled, modular pipeline.
- Exploring the effects of PCA thresholding and channel selection.
- Evaluating per-category performance and generalization.
- Testing domain transfer and synthetic anomaly generation for categories with limited defect samples.

2.7 Broader Applications

Beyond manufacturing, deep learning-based anomaly detection has shown great potential across various domains:

- **Oil and Gas:** Detecting pipeline leaks, corrosion, pressure anomalies, and structural faults in refineries or offshore platforms using vision-based or sensor fusion techniques.
- **Industrial Infrastructure:** Monitoring bridges, tunnels, and machinery for early signs of damage or wear.
- **Medical Imaging:** Identifying tumors, lesions, or other abnormalities in X-rays, MRIs, and histopathological images.
- **Smart Cities & Transportation:** Detecting cracks or surface defects on roads, rails, and airport runways.
- **IoT & Cybersecurity:** Spotting abnormal patterns in system logs, network activity, or connected device behavior.

3. Methodology

This section outlines the complete pipeline used for unsupervised image anomaly detection, including data preprocessing, feature extraction, statistical modeling, and anomaly scoring. The overall framework is inspired by PaDiM but includes several customizations and ablations to evaluate different backbone architectures and hyperparameter configurations. Figure 1 illustrates the overall anomaly detection pipeline implemented in this project, combining key components such as feature extraction, PCA, Gaussian modeling, and Mahalanobis scoring.

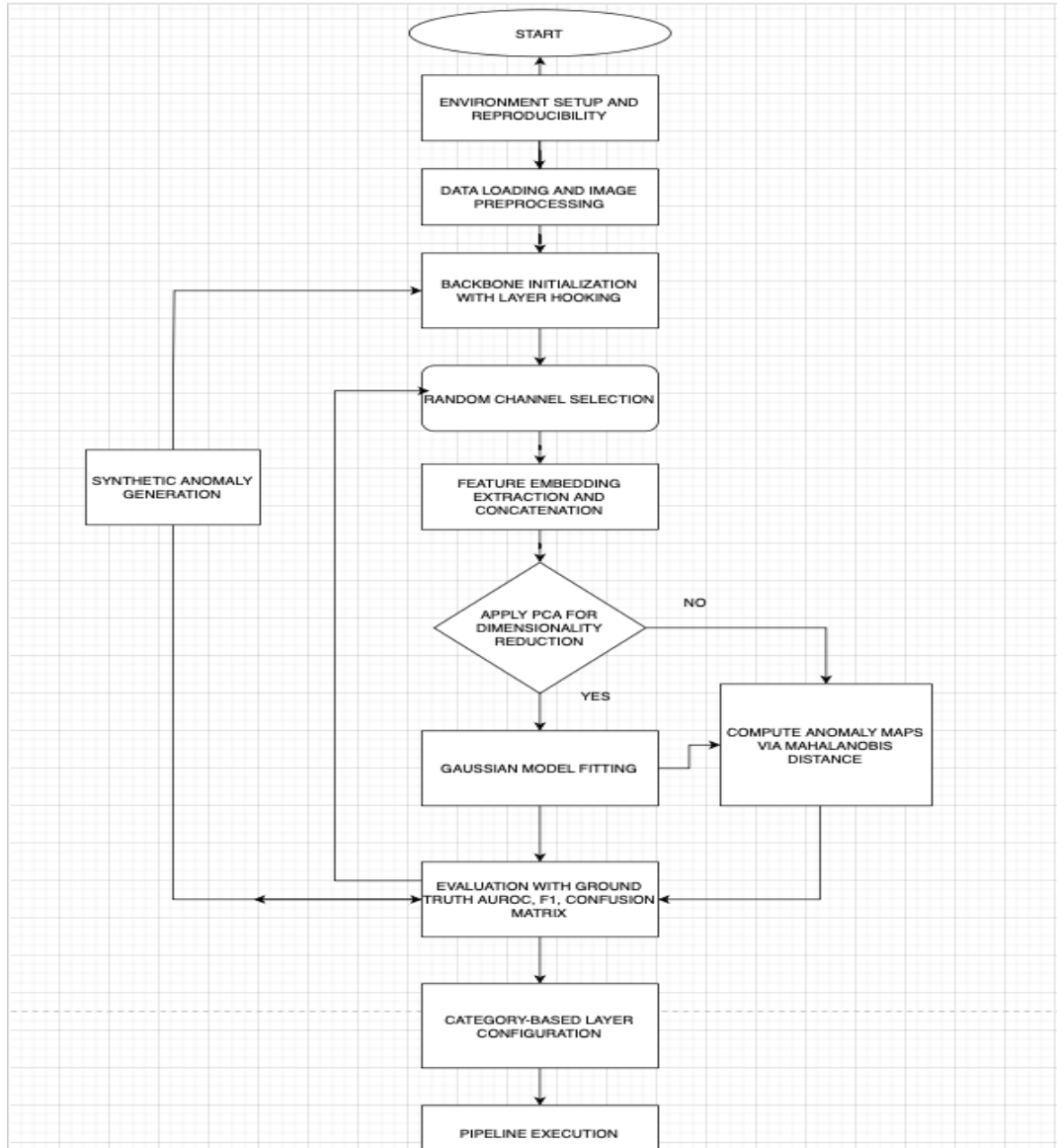


Figure 1. Anomaly detection pipeline architecture inspired by PaDiM with custom modules for synthetic anomaly generation, category-specific tuning, and PCA-based scoring.

Assumptions

1. This project assumes that all training images used are completely defect-free. This is important because the whole approach depends on learning what “normal” looks like, so even a few unnoticed defects in the training set could distort the distribution and reduce the model’s ability to correctly identify anomalies later.
2. It’s also assumed that the feature distributions of normal samples won’t shift much between training and testing. This assumption allows us to use a single learned Gaussian model Throughout, without needing to adapt it per test image.
3. Features at each patch location are treated independently. That means we don’t model any spatial relationship between nearby patches. While this simplifies things, it also means we rely heavily on the individual patch statistics.
4. PCA is assumed to keep the most important information needed to tell normal and abnormal regions apart. It’s also assumed that the number of PCA components chosen will work well across different categories without major loss of information.
5. For underrepresented classes (like toothbrush or transistor), synthetic anomalies are introduced to improve model balance. Here, it’s assumed that the generated defects are meaningful enough to help the model generalize better without introducing unwanted noise or misleading patterns.

3.1 Dataset and Preprocessing

We use the publicly available MVTec Anomaly Detection (MVTec-AD) dataset, which contains 15 industrial object and texture categories. Each category includes a small set of normal training images and a larger test set comprising both normal and defective images, with corresponding ground truth masks for pixel-level evaluation. Images are resized and normalized using either ImageNet statistics (mean and std) or dataset-specific computed values. Augmentations such as random horizontal flips are applied only during training. A configurable transformation pipeline allows flexible tuning of input size and normalization parameters.

3.1.1 Mask Handling

Defect segmentation masks for "defective" images are loaded for pixel-level evaluation. These ground truth masks are essential for calculating pixel-level AUROC and for generating visual heatmap overlays that compare predicted anomaly regions with actual defect locations. The masks are binary and aligned spatially with input images, enabling precise evaluation of detection localization performance.

3.2 Feature Extraction via CNN Backbones

Several pretrained convolutional neural networks (CNNs) are evaluated as feature extractors, including:

- **ResNet18** and **ResNet50** (residual networks) Feature residual connections for effective deep representation learning.
- **DenseNet121** (dense connectivity): Employs dense connections for feature reuse, beneficial for fine-grained localization.
- **EfficientNet-B5** (compound scaling): Compound scaling optimizes depth and parameter efficiency.

Intermediate feature maps are extracted from selected blocks or layers. We experimented with:

- Using all the blocks
- Using only the last block of each stage (to reduce noise and emphasize deep semantic features).
- Concatenating features from both shallow and deep layers (to retain fine-grained and global structure).

A feature hook mechanism is implemented to extract patch-wise embeddings for each training image, preserving spatial positions.

3.3 Dimensionality Reduction using PCA

The extracted features from normal images are optionally compressed using Principal Component Analysis (PCA) with configurable output dimensions. PCA serves to:

- Reduce redundancy and computational cost,
- Denoise features for better statistical modeling,
- Improve anomaly localization by focusing on dominant directions of variation.

We experimented with varying PCA dimensions (e.g.90,95,98,99) and observed that optimal settings vary across categories.

3.4 Gaussian Modeling of Normal Features

After extracting features from the selected backbone model and optionally reducing dimensionality with PCA, the next step in the pipeline is to statistically model the distribution of “normal” features at each spatial patch location. This is essential for establishing a reference distribution against which anomalies in test images can be detected.

We model the distribution of features across all training images as a multivariate Gaussian distribution at each spatial location (i,j) , based on the normal (non-defective) training samples. This results in a location-specific mean vector and covariance matrix that define what is “expected” at each patch.

Mean Vector

As shown in Equation (1), the mean vector μ_{ij} at a patch location (i,j) summarizes the average expected feature values from all normal training images:

$$\mu_{ij} = \frac{1}{N} \sum_{K=1}^N x_{ij}^{(K)} \dots\dots\dots(1)$$

To capture the variability and relationships between patch features, the covariance matrix is computed. The covariance in Equation (2) enables Mahalanobis-based scoring by modeling both spread and correlation at each location:

$$\Sigma_{ij} = \text{Covariance}(\{x_{ij}^{(k)}\}_{k=1}^N) \dots\dots\dots(2)$$

3.5 Anomaly Scoring via Mahalanobis Distance

During inference, feature vectors are extracted from test images using the same CNN backbone. At each patch location, a Mahalanobis distance is computed between the test feature and the corresponding Gaussian model from training. This results in:

$$M(x) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

- A patch-wise anomaly map,
- A single image-level anomaly score, defined as the maximum value in the anomaly map.

Both pixel-level and image-level scores are used for evaluation.

3.6 Evaluation Metrics

We use the following metrics for performance evaluation:

- AUROC (Area Under ROC Curve) at image and pixel levels,
- Precision, Recall, and F1-score,
- Confusion matrix components: True Positives (TP), False Positives (FP), etc.

Thresholds are either selected by maximizing F1 or via fixed percentile-based heuristics.

3.7 Synthetic Anomalies and Domain Transfer

To test generalization and robustness, we introduced:

- Synthetic anomalies using random perturbation masks on normal images,
- Domain transfer by applying models trained on one category to predict defects in another.

This allowed evaluation of overfitting (especially for small-sample classes like toothbrush and transistor) and potential for cross-category generalization.

3.8 Category-Specific Adaptation and Pipeline Flexibility

- **Layer Configuration:** The pipeline supports per-category customization of block/layer configuration, reflecting empirical optimization for varying defect types (surface, screw, complex objects).
- **Parameter Modularity:** All hyperparameters (e.g., number of channels, PCA dimensions, feature blocks) are exposed for grid or manual tuning.

3.9 Hyperparameter Tuning

We performed controlled experiments to study the impact of:

- Different backbones,
- PCA output size,
- Number and type of feature blocks used,
- Channel selection strategies (random vs. fixed top-k),
- Input resolution and patch size.

3.10 Evaluation Metrics

- To assess the effectiveness of the anomaly detection model, we used a combination of quantitative and qualitative evaluation metrics.
- **AUROC**
The Area Under the Receiver Operating Characteristic Curve (AUROC) measures the model's ability to distinguish between normal and anomalous samples. A higher AUROC indicates better discrimination and is particularly useful in imbalanced datasets.
- **Confusion Matrix**
The confusion matrix summarizes prediction results by comparing ground truth with predicted labels. It helps identify true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), offering a clearer view of classification performance.
- **Precision / Recall**
Precision indicates the proportion of true anomalies among predicted anomalies, while recall measures how many actual anomalies were correctly identified.
- **Qualitative Inspection (Good vs Defect)**
Visual inspection of heatmaps and predicted masks allows us to assess how well the model localizes defects. This step is crucial for interpreting model decisions and validating detection quality beyond numeric scores.

4.RESULTS AND EVALUATION METRICS

Table 1 presents the AUROC scores across 15 categories in the MVTec-AD dataset, evaluated using four different CNN backbones: ResNet18, ResNet50, DenseNet121, and EfficientNet-B5. Each model was trained exclusively on normal images and tested on a mix of normal and defective samples.

From the results in Table 1, EfficientNet-B5 consistently outperformed other backbones in most categories, particularly in complex defect types such as *zipper*, *transistor*, and *tile*. ResNet50 also achieved strong results, especially in texture-rich categories like *wood* and *carpet*. ResNet18 generally had the lowest performance, with significant drops in categories such as *pill* (AUROC = 80.31). DenseNet121 performed competitively, especially in categories requiring fine-grained localization.

Table 1. AUROC Scores for Each Backbone Model across MVTec-AD Categories

CATEGORIES	RESNET18	RESNET50	DENSNET	EFFICIENET
Screw	85.45	91.56	87.19	96.31
Bottle	99.21	99.44	99.84	99.74
pill	80.31	90.51	87.56	91.84
zipper	88.87	90.89	96.98	98.87
wood	96.27	99.47	95.53	95.53
transistor	98.38	98.42	91.50	98.58
toothbrush	97.50	96.11	99.44	96.39
tile	95.13	98.63	87.30	97.91
Metal_nut	96.28	96.97	92.96	97.41
leather	100	100	98.44	100
hazelnut	93.68	94.75	94.00	91.21
grid	86.80	94.74	98.58	92.73
capsule	88.75	89.91	93.78	86.76
cable	91.30	95.39	96.33	96.48
carpet	96.67	99.48	98.48	99.36

4.1. Category-wise AUROC Performance

Table 1 compares the performance of all four backbones i.e. ResNet18, ResNet50, DenseNet121, and EfficientNet-B5 across all categories. From the table, EfficientNet consistently outperformed other models in most categories, especially for challenging classes like *zipper*, *transistor*, and *tile*, achieving AUROC values above 98%. DenseNet showed strong performance in categories with complex textures like *wood* and *capsule*. However, ResNet18 lagged behind in most categories, especially in *pill*, where it recorded an AUROC of just 80.31%.

4.2. Image-Level Evaluation Metrics (Based on AUROC)

True Positive (TP): The model correctly detects an anomaly where there is actually an anomaly in the ground truth.

False Positive (FP): The model detects an anomaly where there is none in the ground truth.

True Negative (TN): The model correctly predicts a normal area where the ground truth also shows no anomaly.

False Negative (FN): The model misses an actual anomaly, it says the area is normal even though it's defective in the ground truth.

Below is a breakdown of the image-level results for some category, showing the best-performing model (based on AUROC) along with the associated metrics:

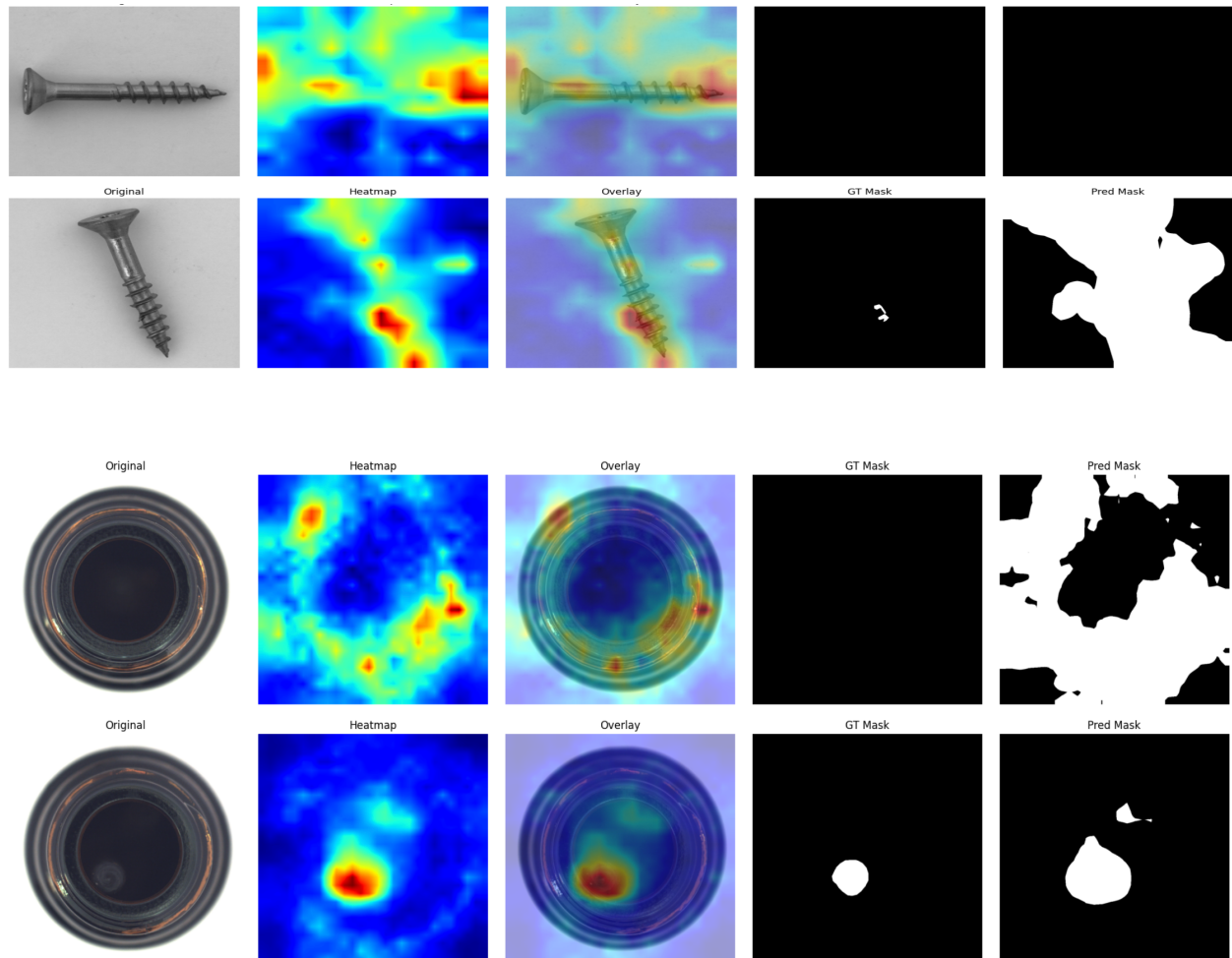
- **Screw (ResNet50, AUROC: 91.56%)**
 - TP: 115, FP: 11, TN: 30, FN: 4
 - Precision: 0.9127, Recall: 0.9664
- **Bottle (EfficientNet, AUROC: 99.76%)**
 - TP: 63, FP: 1, TN: 19, FN: 0
 - Precision: 0.9844, Recall: 1.0000
- **Pill (EfficientNet, AUROC: 91.84%)**
 - TP: 134, FP: 8, TN: 18, FN: 7
 - Precision: 0.9437, Recall: 0.9504
- **Zipper (EfficientNet, AUROC: 98.87%)**
 - TP: 116, FP: 2, TN: 30, FN: 3
 - Precision: 0.9831, Recall: 0.9748
- **Wood (ResNet50, AUROC: 99.47%)**
 - TP: 60, FP: 7, TN: 12, FN: 0
 - Precision: 0.8955, Recall: 1.0000
- **Transistor (EfficientNet, AUROC: 98.58%)**
 - TP: 36, FP: 1, TN: 59, FN: 4
 - Precision: 0.9730, Recall: 0.9000
- **Toothbrush (ResNet18, AUROC: 97.50%)**
 - TP: 30, FP: 4, TN: 8, FN: 0
 - Precision: 0.8824, Recall: 1.0000
- **Tile (EfficientNet, AUROC: 97.91%)**
 - TP: 81, FP: 1, TN: 32, FN: 3
 - Precision: 0.9878, Recall: 0.9643
- **Metal Nut (EfficientNet, AUROC: 97.41%)**
 - TP: 89, FP: 2, TN: 20, FN: 4
 - Precision: 0.9780, Recall: 0.9570

4.3. Backbone Comparison Summary

- EfficientNet-B5 was overall the most reliable and robust model across nearly all categories, achieving the highest average AUROC and consistently high precision and recall.
- ResNet50 followed closely and showed stability across simple and textured surfaces.
- DenseNet121 provided good results for texture-heavy categories like wood and capsule but required careful tuning.
- ResNet18, being a shallower model, struggled in complex anomaly types but still achieved perfect recall in some smaller categories.

4.4. Qualitative Evaluation: Heatmaps & Localization

Shows qualitative results of the anomaly detection pipeline, highlighting the model's localization of defects via heatmaps and binary masks.



In the “Bottle” image:

- The anomaly heatmap and overlay produced by EfficientNet highlights the true defect region.
- The predicted mask has high spatial overlap with the ground truth mask, with minimal background noise.

In the “Screw” image (good sample):

- The anomaly score heatmap appears nearly uniform and low-intensity, matching the lack of defects the predicted mask is blank, accurately reflecting a normal instance.

4.5 Error Characteristics

- False Positive Examples: Almost absent in object categories, but slightly higher in highly textured or naturally variable backgrounds (grids, wood).
- False Negatives: Occasionally encountered for extremely subtle or tiny defects, or for types rare in the training set.

4.6. DISCUSSION

The results demonstrate that deeper feature extractors such as DenseNet121 and EfficientNet-B5 consistently outperform shallower models like ResNet18, particularly in categories that require nuanced understanding of complex textures (e.g., leather, carpet, metal nut). However, this gain in accuracy comes with trade-offs. Simply using all blocks or deeper layers doesn't guarantee better performance tuning was essential, as extracting features from noisy or redundant layers increased false positives. Optimal performance was typically achieved when features were extracted from later, more abstract layers.

Synthetic data generation helped test model generalization, particularly for underrepresented categories like *toothbrush* and *transistor*. Basic augmentation methods such as random masking was used due to hardware constraints (i.e, lack of CUDA support), but the results indicate that higher-quality synthetic data is necessary. Poorly designed synthetic defects introduced noise, which degraded performance, highlighting the importance of more advanced techniques like GANs or interpolation-based methods (i.e, AnomalyAny).

The domain transfer experiment showed that features learned from one category (e.g., cable) can detect anomalies in a visually similar but unseen category (e.g., grid), suggesting some shared representation learning across categories. However, category-specific tuning still yielded the best results, supporting the notion that domain generalization in unsupervised anomaly detection remains challenging.

Implications and Industrial Relevance

The findings in this study demonstrate several real-world benefits across multiple industrial sectors:

- Oil & Gas: High-performing models (AUROC > 95% in many categories) suggest the pipeline can be deployed for pipeline monitoring, equipment health monitoring, visual asset inspection, and subsurface imaging. This supports proactive maintenance and environmental risk reduction.
- Manufacturing: The model's success in identifying subtle surface anomalies (e.g., screws, capsules, tiles) points to applications in quality control, automated defect inspection, and production speed optimization without human intervention.
- Pharmaceutical and Food Industries: Categories like *bottle* and *pill* achieved high accuracy, indicating that anomaly detection can be used to ensure safety and hygiene during packaging and large-scale production.
- Automotive and Aerospace: The ability to localize small, irregular defects supports use in component assembly validation and compliance with strict safety standards.

5. Conclusion

5.1. Summary of Findings

This project successfully implemented an anomaly detection pipeline using patch-wise Mahalanobis scoring on features extracted from different CNN backbones. The method was tested on the MVTec-AD dataset, with evaluation based on both image-level AUROC and pixel-level localization results. The model was able to produce reliable heatmaps and predicted masks that closely aligned with ground truth, especially in categories like hazelnut, screw, and bottle. Evaluation metrics such as precision, recall, F1-score, and confusion matrix showed strong performance, particularly when using deeper backbones. Qualitative visualizations confirmed that the model could detect subtle and small defects effectively.

5.2. Best Performing Backbone

Among all tested architectures, **EfficientNet-B5** consistently outperformed others in terms of both detection accuracy and anomaly localization. It produced more concentrated heatmaps, better mask overlap with ground truth, and higher AUROC scores. Compared to ResNet18, ResNet50 and DenseNet121, EfficientNet-B5 extracted richer and more discriminative features, leading to improved detection across both texture and structural categories.

5.3. Future Work

- **Domain Transfer**
work will involve domain transfer evaluation using real images from pipelines, rigs, and equipment inspections.
- **Lightweight Deployment**
EfficientNet-B5 gave the best accuracy, but it is large and slow for deployment in real-time settings. Future work can involve experimenting with compressed models or low-parameter networks to enable real-time monitoring on low-resource devices.
- **Synthetic Anomaly Injection**
For underrepresented classes with few defects, synthetic data generation can help balance the dataset. Future work can explore GAN-based methods or interpolation techniques like AnomalyAny to boost model learning without collecting new data.
- **Real-Time System Potential**
With improvements in processing speed and efficient model designs, this pipeline can be adapted into a real-time anomaly detection system for industrial visual inspection. This would allow direct deployment in production facilities or on inspection drones.
- **Vision Mamba-Style Architectures**
Recently developed Mamba-based models are efficient and designed for sequential visual reasoning. Exploring these architectures may improve both speed and long-range pattern understanding in defect detection.
- **Transformer-Based Models**
Transformers like ViT and Swin Transformer have shown success in visual tasks. Their

ability to capture global dependencies could enhance anomaly localization, especially in large or subtle defect regions.

- **GPU Utilization and Parallelism**

Further work can focus on optimizing GPU usage during training and inference.

Techniques like mixed precision training, batch parallelism, or model quantization could help reduce memory usage and boost speed without compromising performance.

- **Adaptive Hyperparameter Tuning**

Manual tuning of PCA, channels, and resizing impacted scalability. Future research can develop auto-tuning strategies or meta-learning techniques to adapt parameters per category or domain.

REFERENCES

- Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.
- Defard, T., Setkov, A., Loesch, A., & Audigier, R. (2021). *PaDiM: A Patch Distribution Modeling Framework for Anomaly Detection and Localization*. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*.
- Deng, Y., & Zhao, Z. (2023). *AnomalyAny: Learning to Generate Any-Shaped Anomalies for Visual Inspection*.
- Roth, K., Moltmann, R., & Lucic, M. et al. (2022). *Towards Total Recall in Industrial Anomaly Detection*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Yi, Z., & Wu, Y. (2023). *GLASS: Global-Local Feature Alignment for Semantic-aware Anomaly Segmentation*.
- Cohen, S., & Hoshen, Y. (2020). *Sub-Image Anomaly Detection with Deep Pyramid Correspondences (SPADE)*. In *BMVC 2020*.
- Bergmann, P., Fauser, M., Sattlegger, D., & Steger, C. (2019). *MVTec AD - A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. In *CVPR 2016*.
- Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. Q. (2017). *Densely Connected Convolutional Networks*. In *CVPR 2017*.
- Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. In *ICML 2019*.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*.
- Chandola, V., Banerjee, A., & Kumar, V. (2009). *Anomaly Detection: A Survey*. *ACM Computing Surveys (CSUR)*.
- Liu, R., Lehman, J., Molino, P., et al. (2020). *A Survey of Deep Anomaly Detection*. *arXiv:2004.06979*.
- Akcay, S., Atapour-Abarghouei, A., & Breckon, T. P. (2018). *GANomaly: Semi-Supervised Anomaly Detection via Adversarial Training*. *ACCV*.
- Bergmann, P., Fauser, M., Sattlegger, D., & Steger, C. (2019). *MVTec AD - A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection*. *CVPR*.
- Defard, T., Setkov, A., Loesch, A., & Audigier, R. (2021). *PaDiM: A Patch Distribution Modeling Framework*. *ICPR*.
- Bergmann, P., Fauser, M., Sattlegger, D., & Steger, C. (2019). *MVTec AD - A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection*. *CVPR*.
- Bergmann, P., Meinhardt, T., & Leal-Taixé, L. (2023). *MVTec AD 2: A Real-World Dataset for Unsupervised Anomaly Detection and Localization*. *arXiv:2302.07360*